

Clinical Risk Assessment Report: Patient 63

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.45x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.3%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.19 [Avg (52th %ile)] (-1.3%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Patient Age (Age) **Value: 68.0 [Avg (68th %ile)] (-1.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.47 [Avg (35th %ile)] (-1.1%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: -0.58 [Bottom 23%] (-0.8%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.53 [Avg (27th %ile)] (-0.7%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.12 [Avg (55th %ile)] (-0.6%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.19 [Avg (35th %ile)] (-0.6%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -0.71 [Bottom 23%] (-0.5%)**
Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Machine Learning Feature Attribution (SHAP Decision Path)

